DATABASES

Laboratory work 6

JOIN queries

06

Student

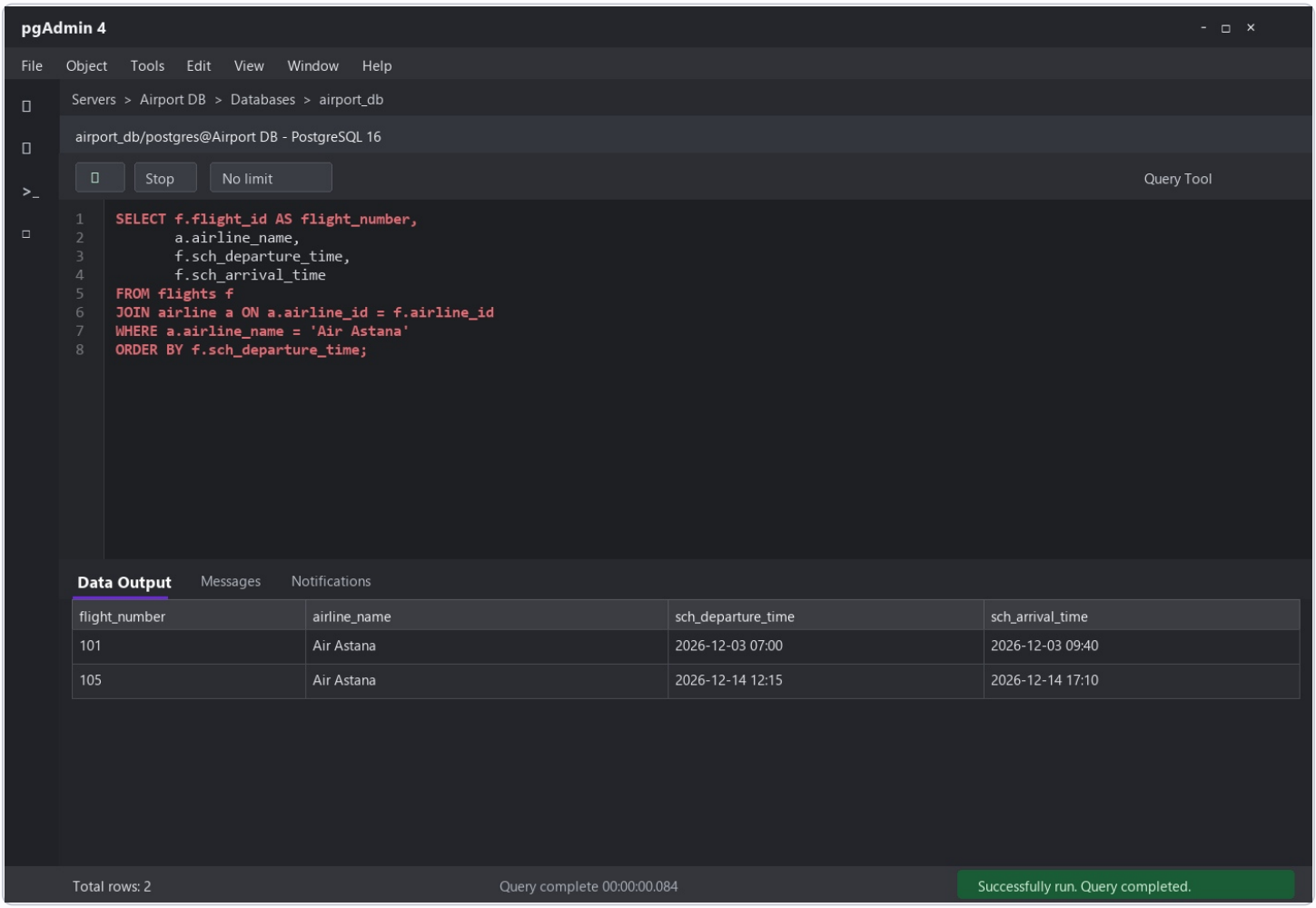
Madi Murat

Database airport_db

PostgreSQL 16

Task 1

Display all flights of a specific airline.

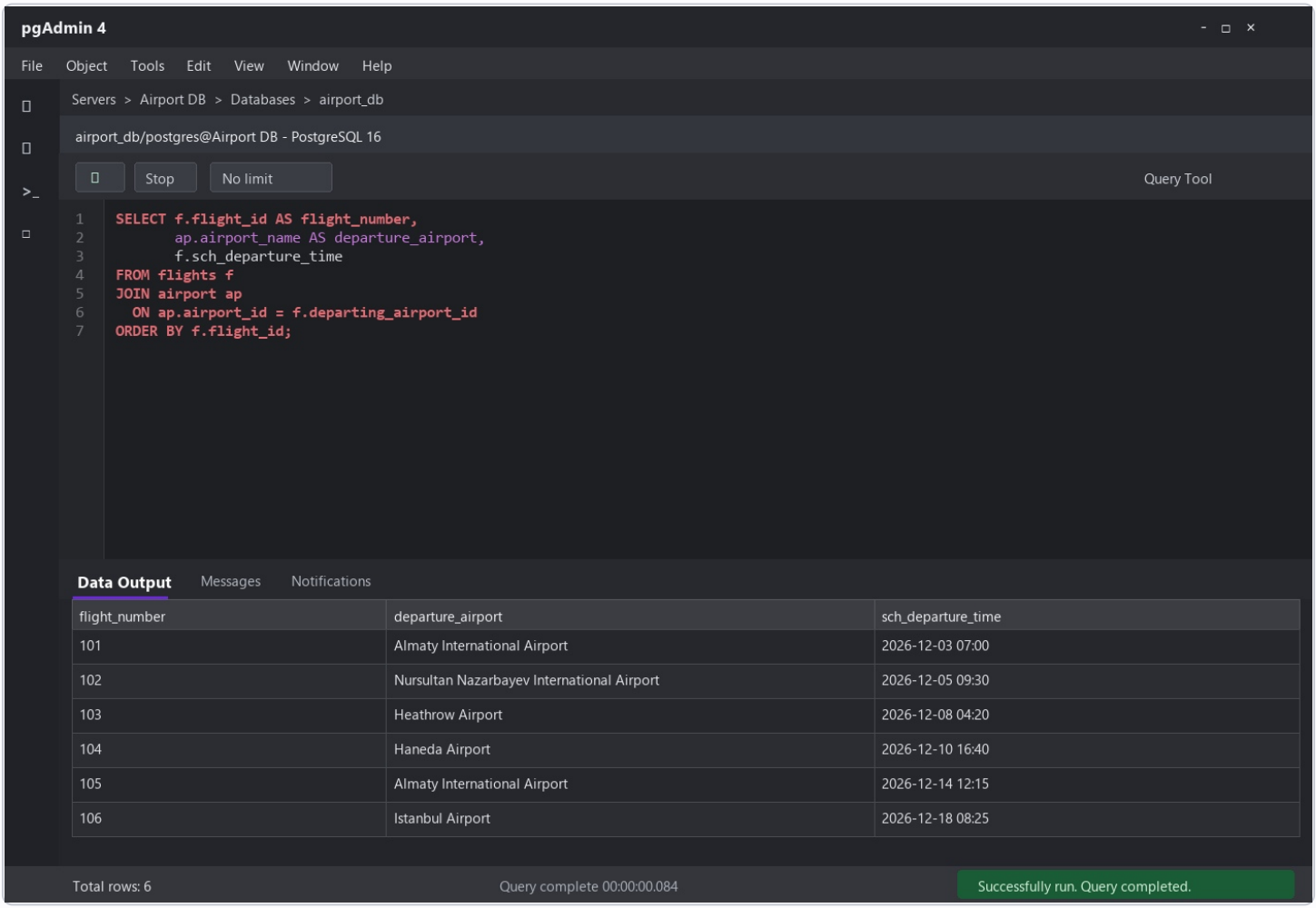


COMMAND EXPLANATION

INNER JOIN connects every flight with its airline through airline_id. WHERE leaves only Air Astana flights, and ORDER BY shows them chronologically.

Task 2

List all flights with departure airport names.

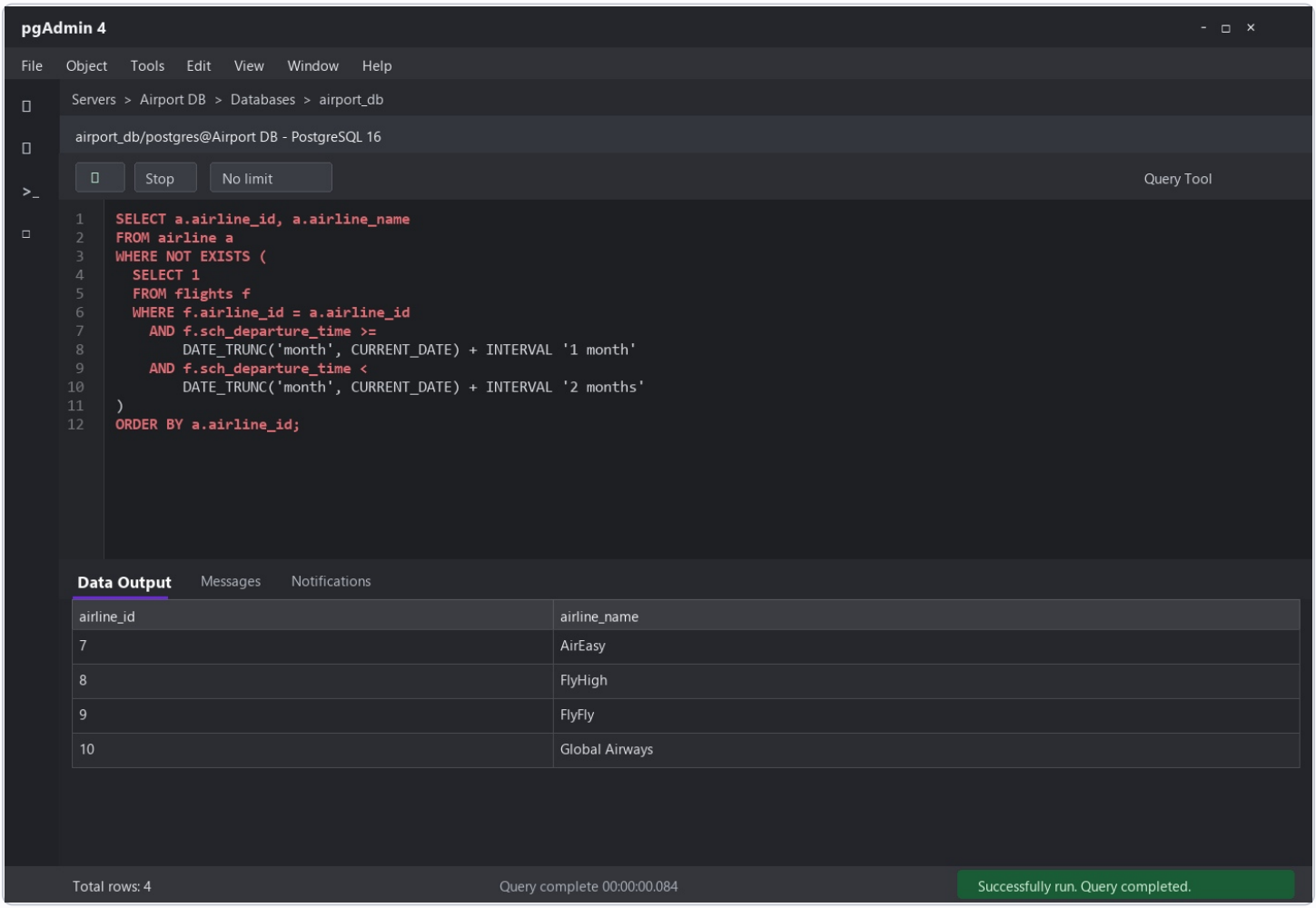


COMMAND EXPLANATION

The join condition matches departing_airport_id with airport_id. The alias departure_airport makes the meaning of the selected airport name clear.

Task 3

Find airlines with no flights scheduled for next month.

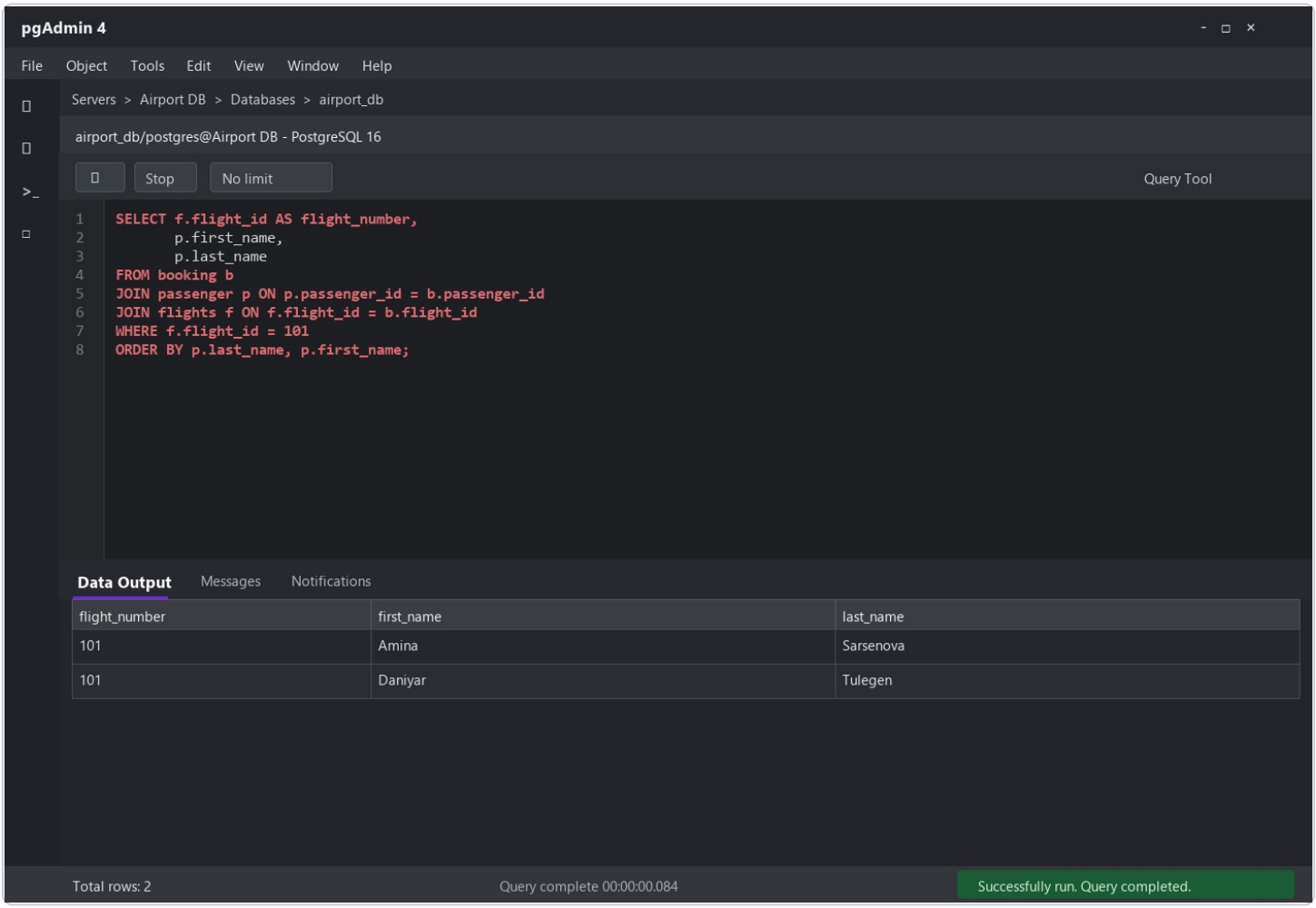


COMMAND EXPLANATION

NOT EXISTS checks every airline against next month's date interval. If no matching flight is found, the airline is included in the result.

Task 4

Display passengers on a specific flight.

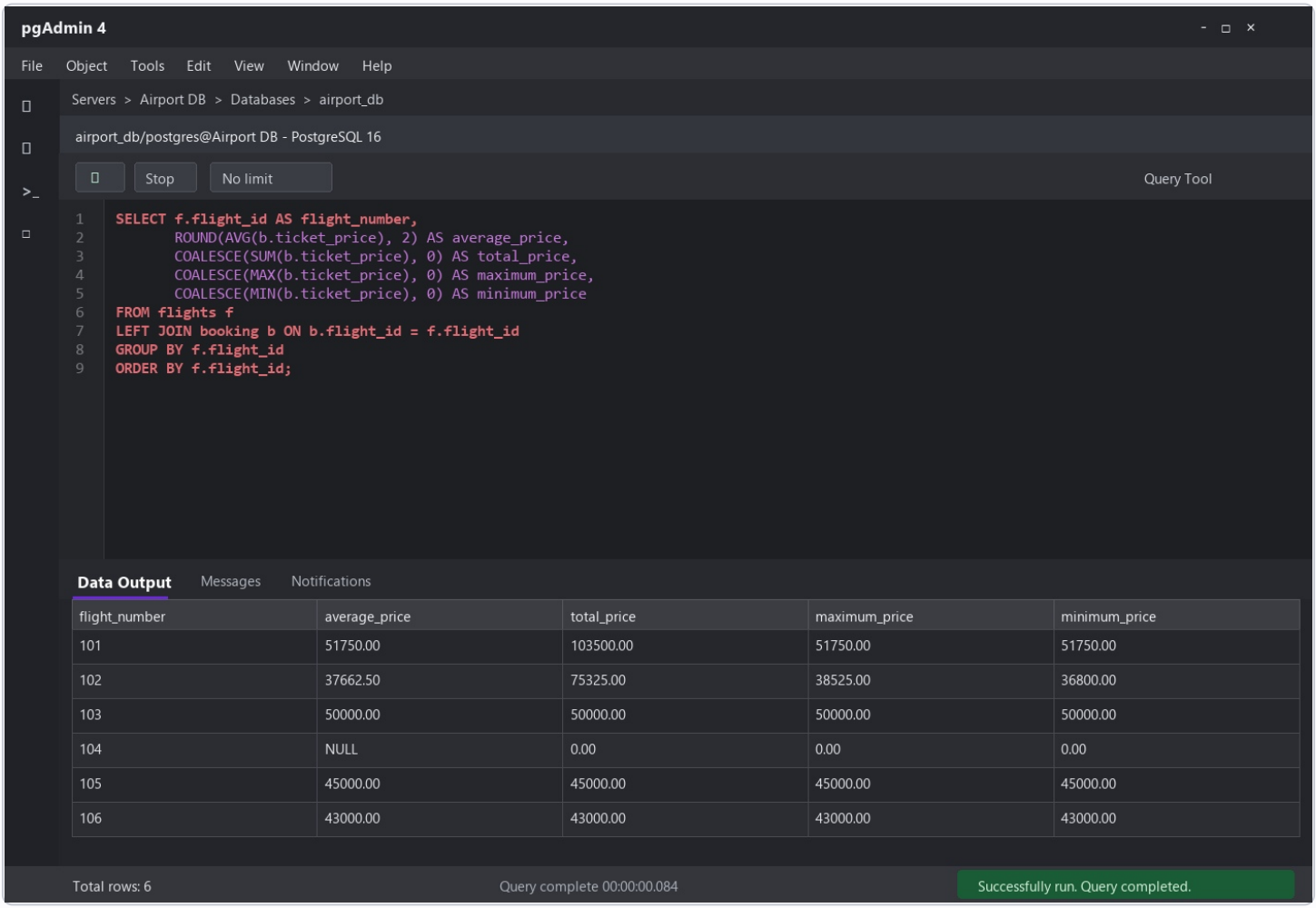


COMMAND EXPLANATION

Booking is the link between passengers and flights. The two joins restore passenger details for flight 101, while ORDER BY sorts the names.

Task 5

Calculate ticket price statistics for each flight.

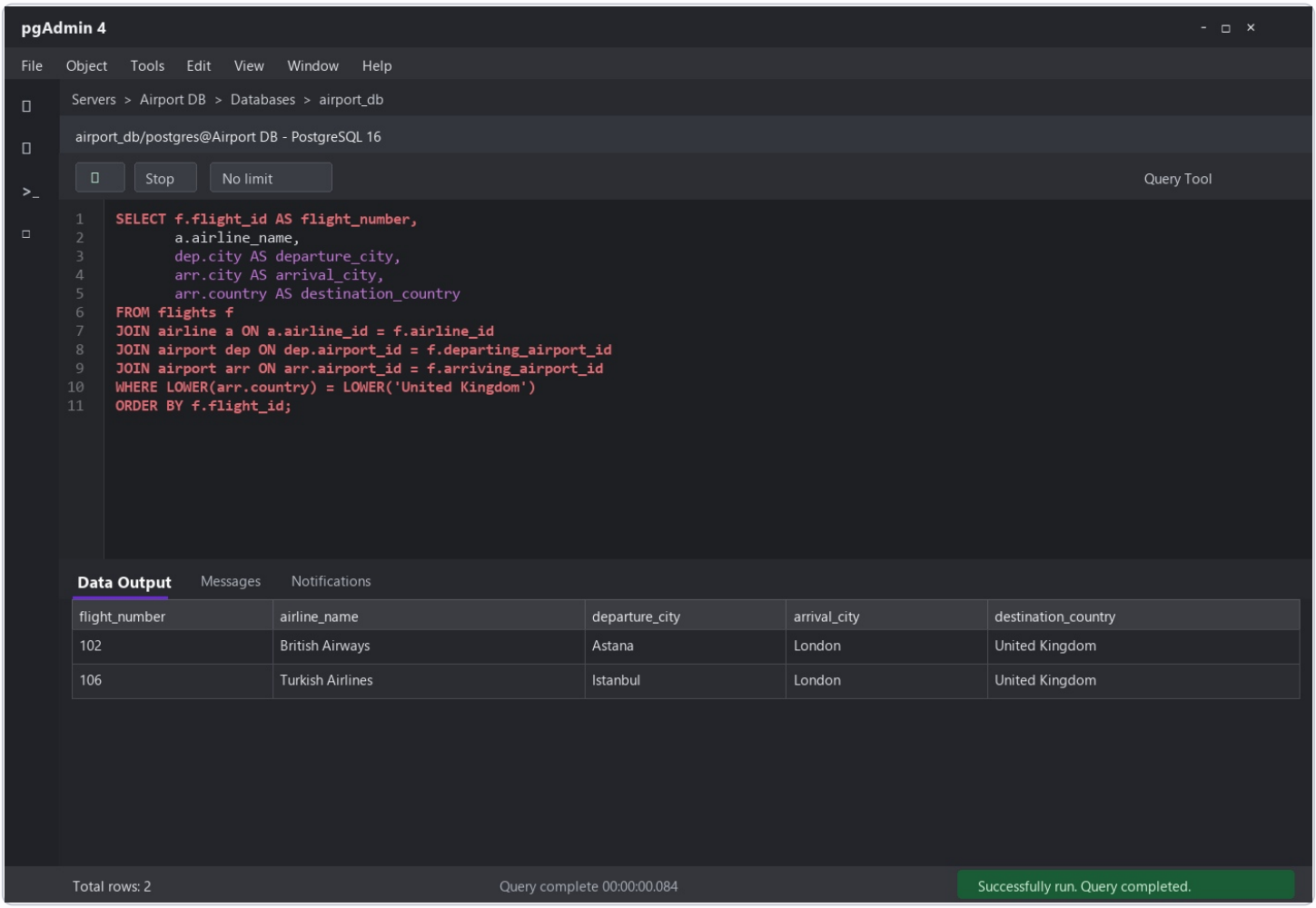


COMMAND EXPLANATION

LEFT JOIN keeps flights that have no bookings. AVG, SUM, MAX and MIN calculate the required values, and COALESCE replaces empty totals and limits with zero.

Task 6

Show flights arriving in a specific country.

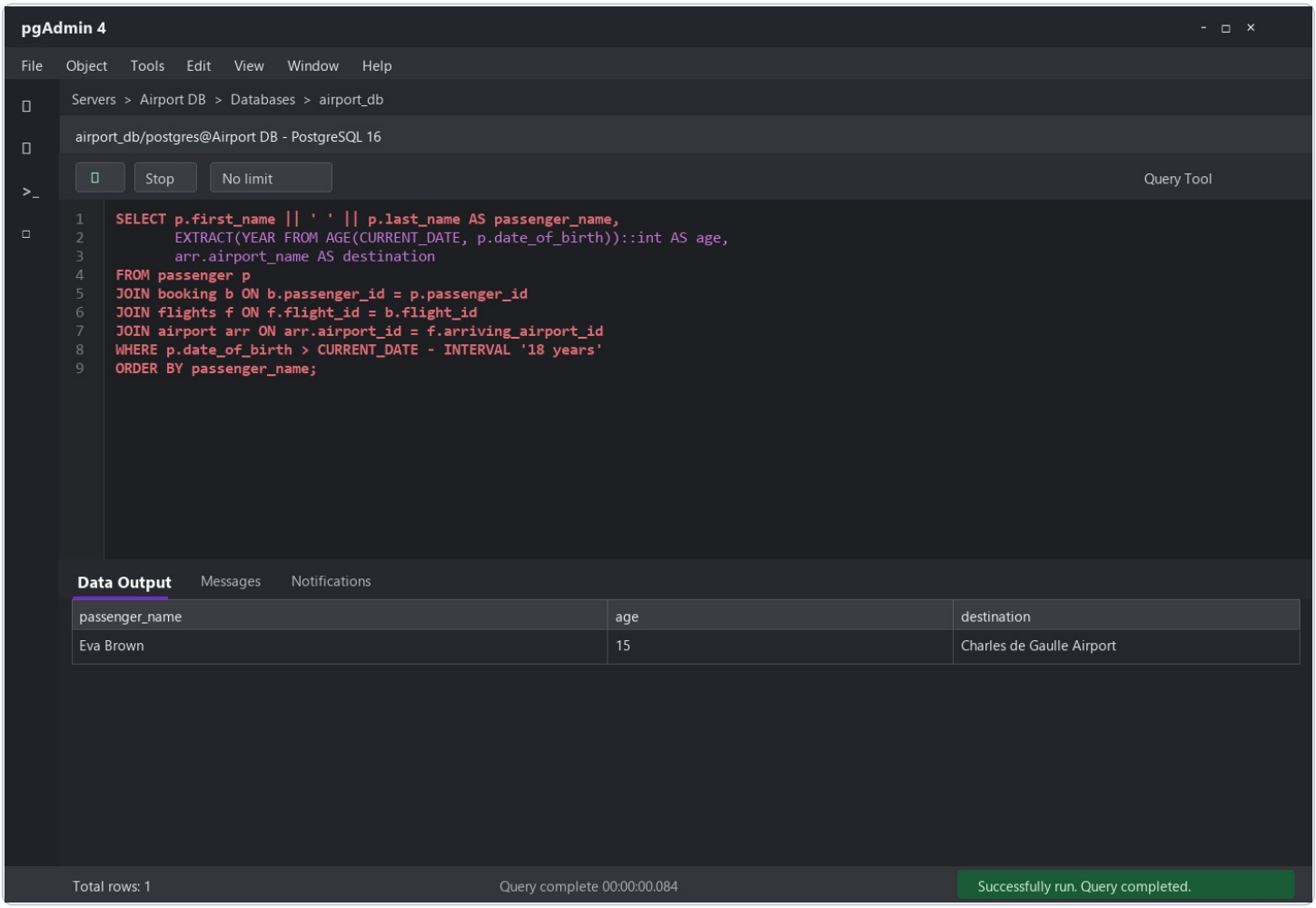


COMMAND EXPLANATION

The airport table is joined twice: dep represents the origin and arr represents the destination. The country condition is applied to arr only.

Task 7

Display minor passengers and their destinations.

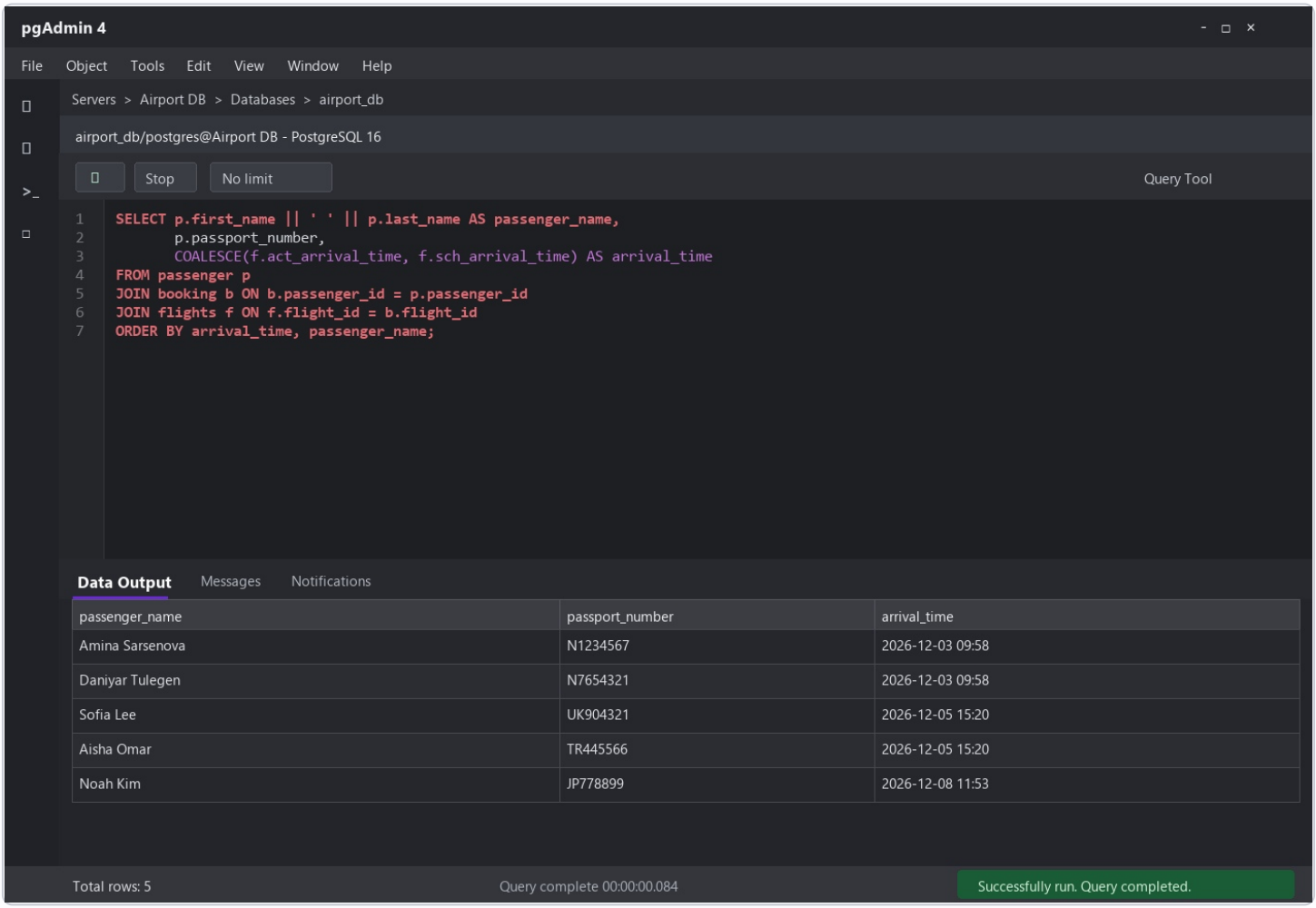


COMMAND EXPLANATION

AGE calculates the passenger's age and the WHERE condition keeps people younger than 18. The arrival airport is reached through booking and flights.

Task 8

Show passenger details and current arrival time.

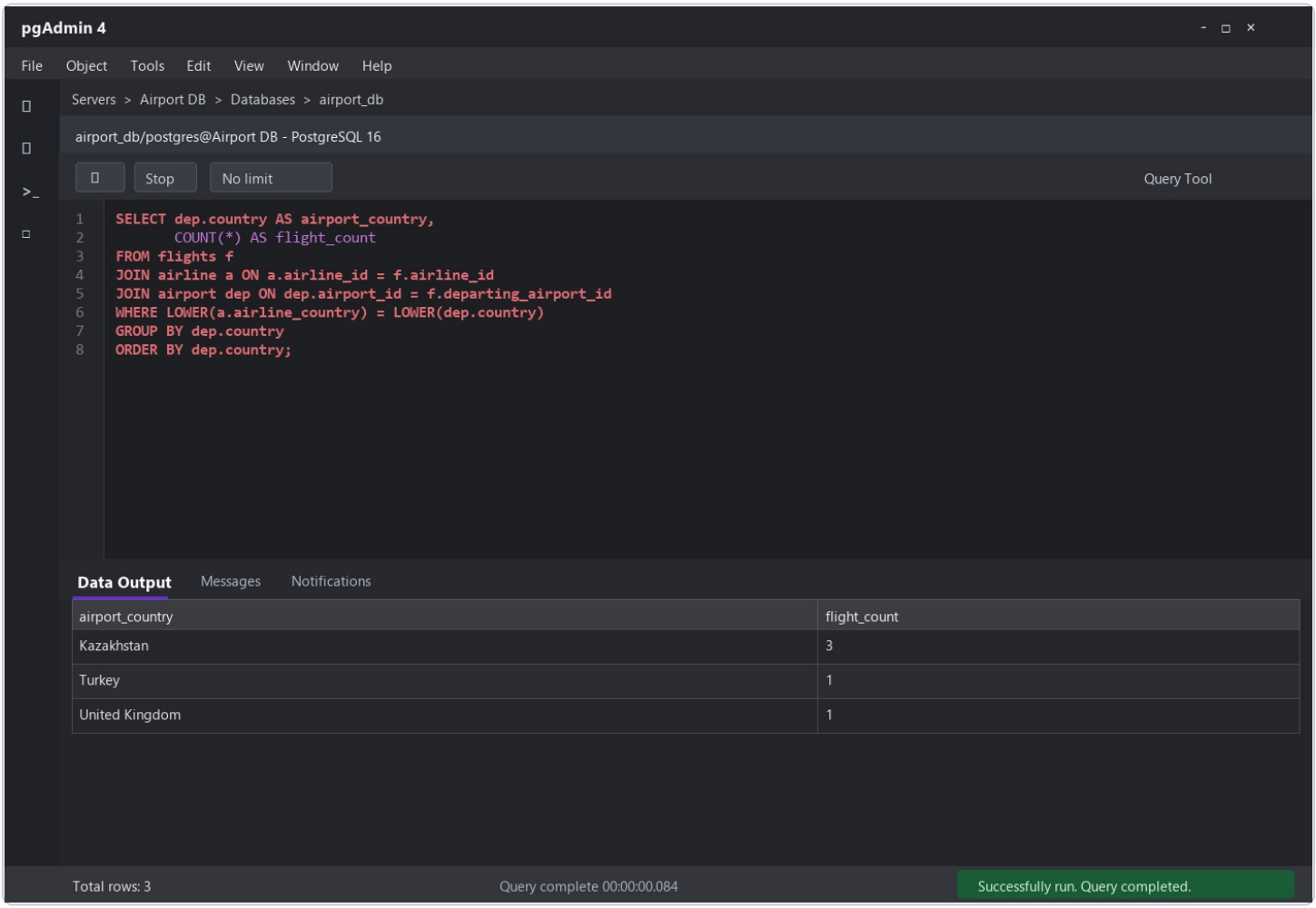


COMMAND EXPLANATION

COALESCE uses the actual arrival time when it exists; otherwise it uses the scheduled time. This keeps the query useful for both completed and future flights.

Task 9

Group domestic-origin flights by airport country.

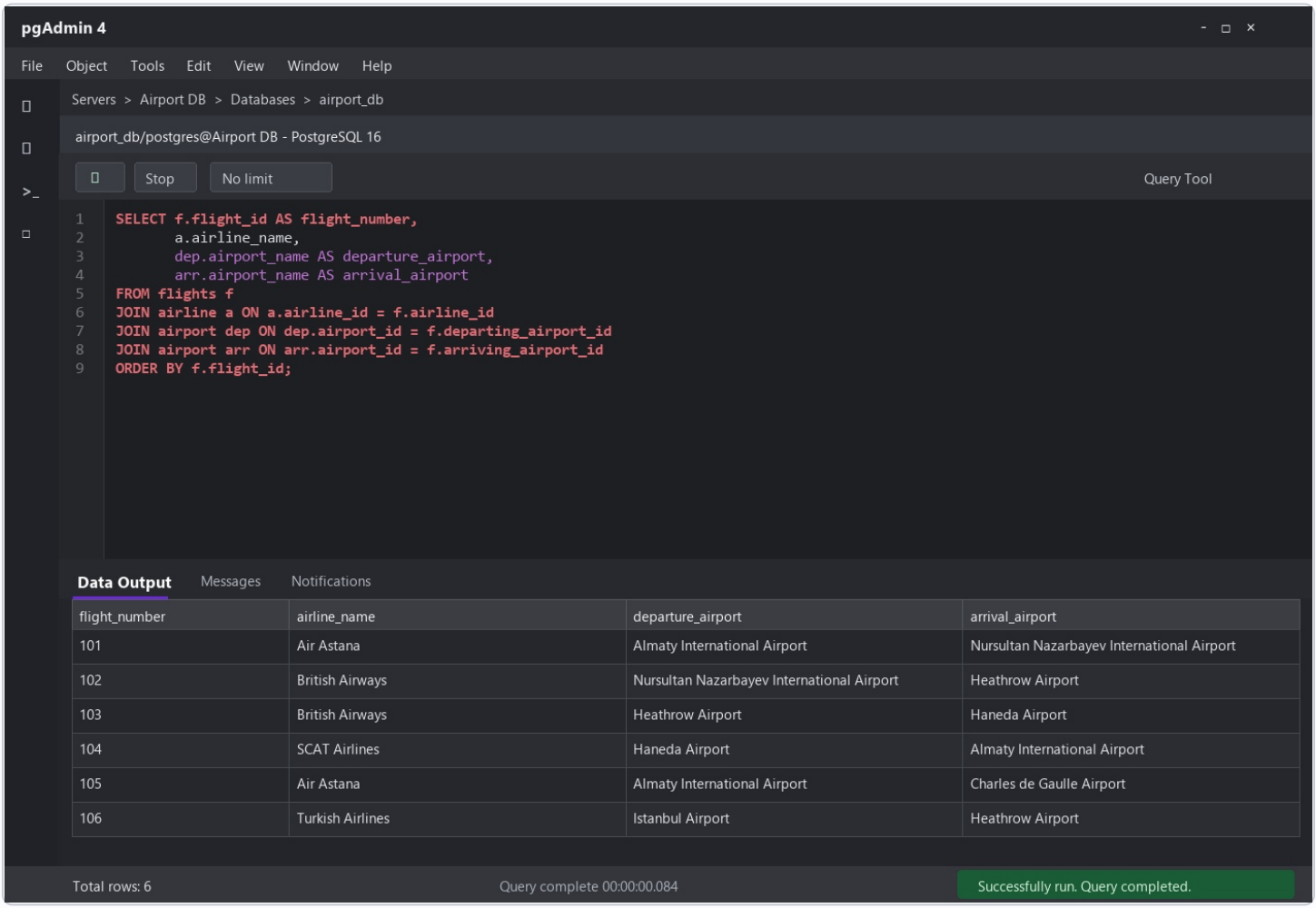


COMMAND EXPLANATION

The WHERE clause compares the airline's home country with the departure airport country. GROUP BY returns one row per country and COUNT gives its number of matching flights.

Task 10

List each flight, airline and both airports.

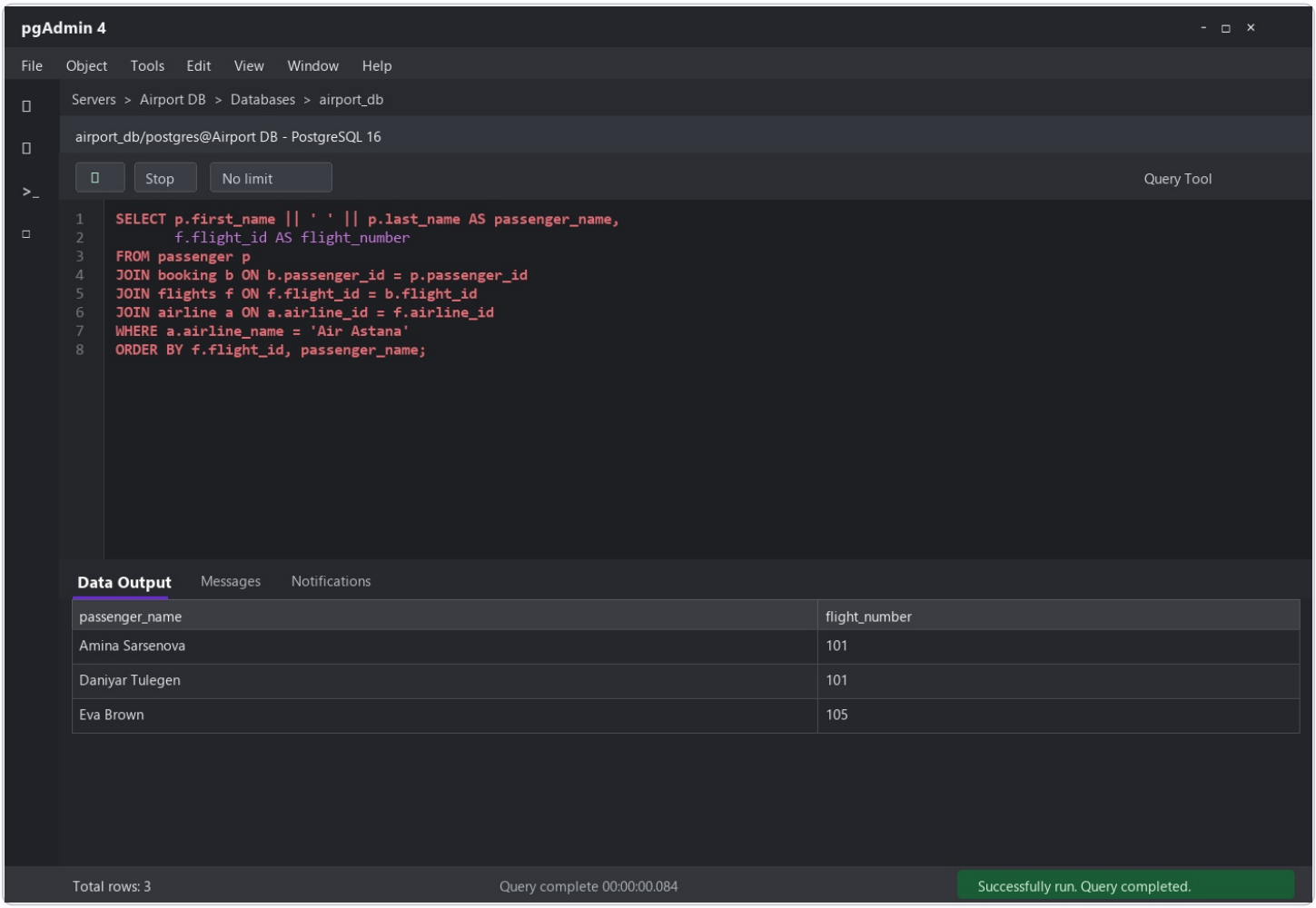


COMMAND EXPLANATION

Flights is joined to airline once and to airport twice. Separate dep and arr aliases prevent ambiguity between departure and arrival airport columns.

Task 11

Find passengers booked with a specific airline.

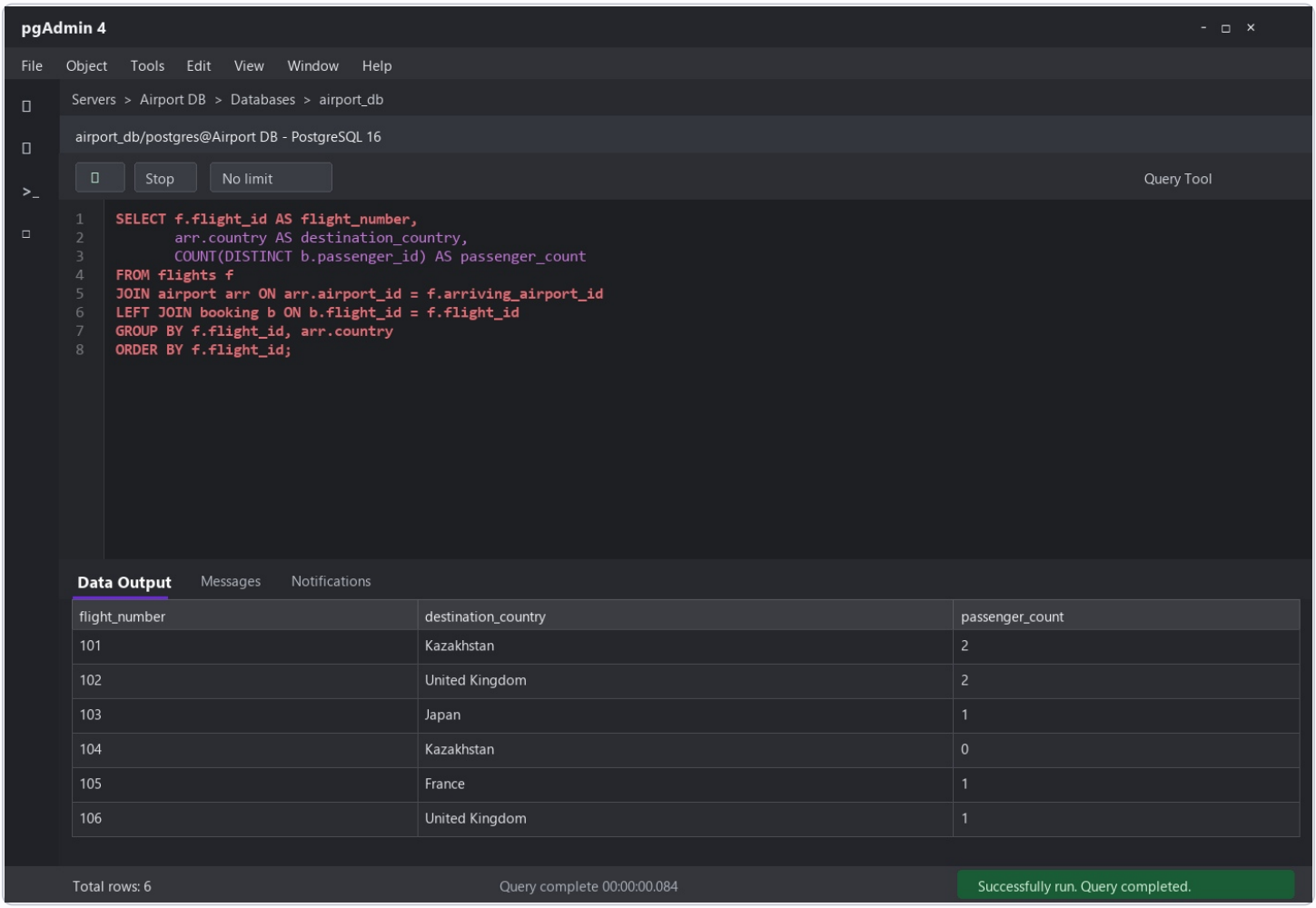


COMMAND EXPLANATION

The join chain follows passenger to booking, flight and airline. The airline name filter is easy to replace with another airline when needed.

Task 12

Count booked passengers for every destination flight.

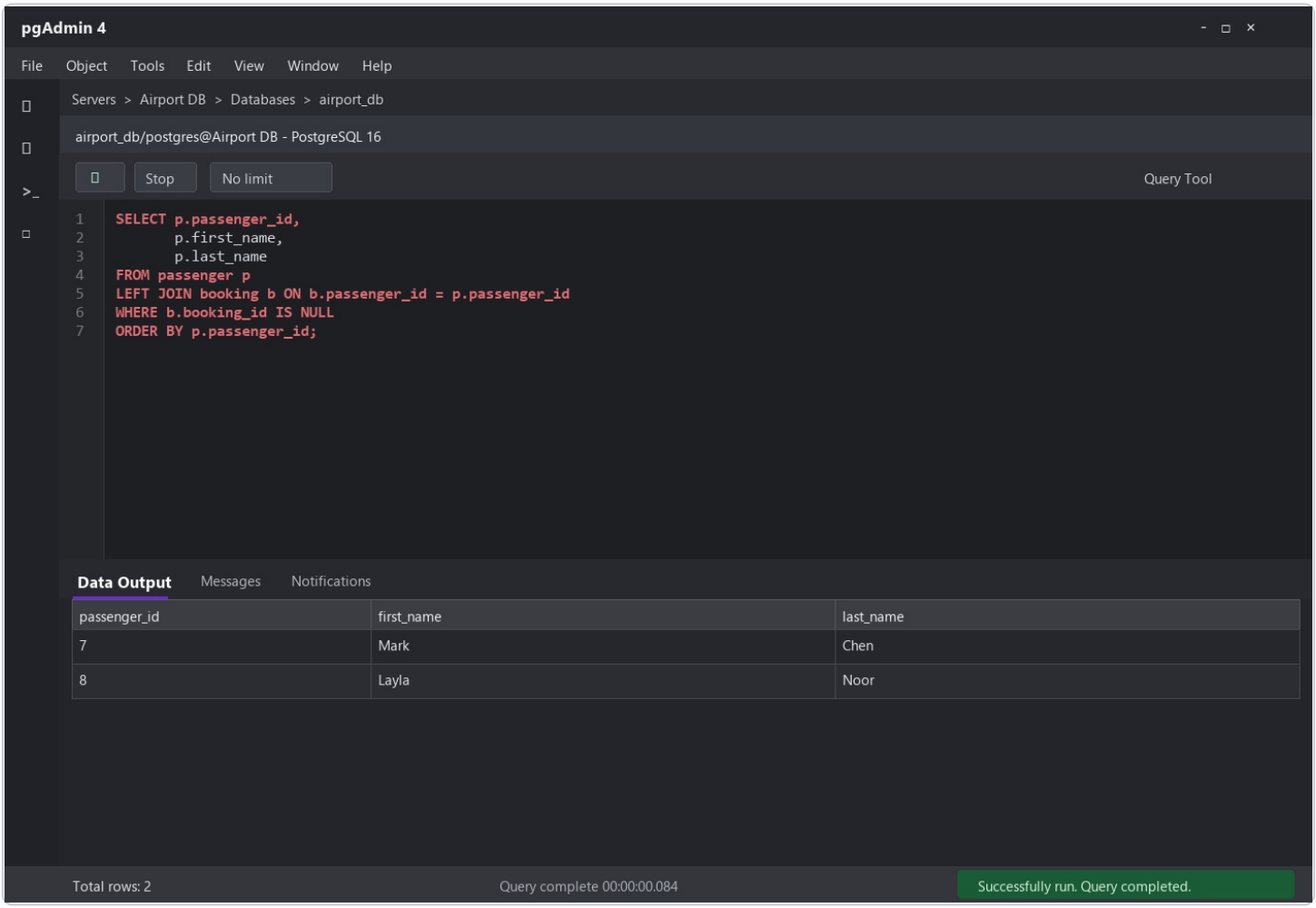


COMMAND EXPLANATION

LEFT JOIN is important because it keeps flights with no bookings. COUNT on passenger_id returns zero for such a flight, while GROUP BY produces one row per flight and destination.

Task 13

Find passengers who have not booked any tickets.

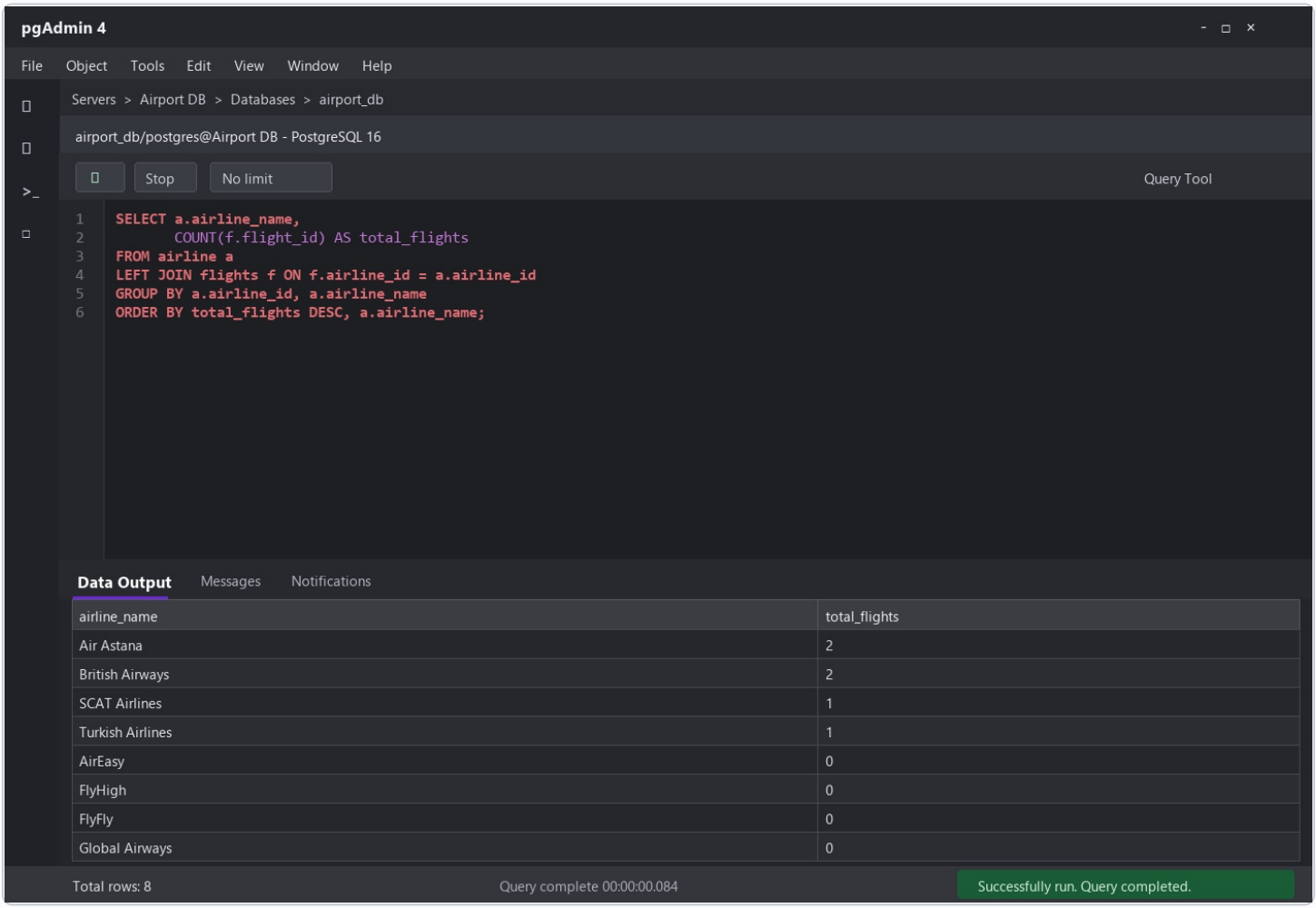


COMMAND EXPLANATION

LEFT JOIN keeps every passenger. A missing booking produces NULL in booking_id, so the WHERE clause identifies passengers without tickets.

Task 14

Count flights for every airline, including zero.

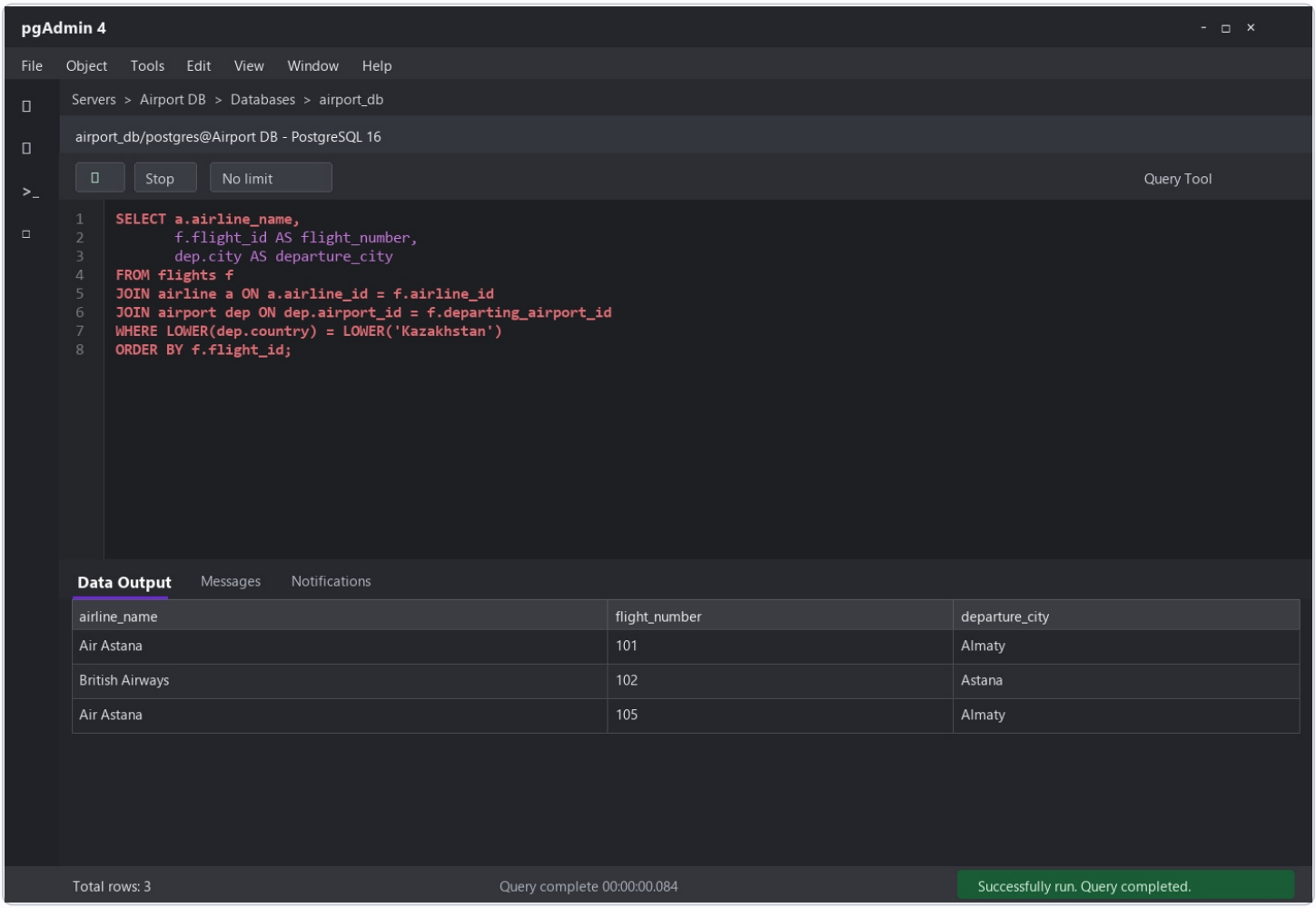


COMMAND EXPLANATION

Airline is placed on the left side of LEFT JOIN so even airlines without flights remain. COUNT(f.flight_id) counts only real flight rows and therefore returns zero when none exist.

Task 15

List flights departing from a specified country.



COMMAND EXPLANATION

The departure airport join provides its city and country. LOWER makes the Kazakhstan filter independent of capitalization in stored data.